ActInf Textbook Group ~ Cohort 7 ~ Session 4 (Chapter 2, part 2) ~ 8/12/2024

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all right welcome back coort 7 we're in our second discussion on chapter two so let's head over to two we can begin with anyone's thoughts or Reflections or any question or idea or section of the text they want to begin with OR curiosity from to or just some quote they want to go to anyone can just raise their hand or type it in the chat okay actual go for it sure so one thing that I found a bit interesting it's on page 26 of the book um it says the person can resolve this discrepancy in two ways it's talking about this thing of looking at um like expecting to see an apple when you look and then actually seeing a frog and it says first she could change her mind about what she's seeing so actually that it's a frog instead of an apple um or she can fiate the nearest apple tree and see something that looks very much like an apple this resolv the initial discrepancy but in a different way um so she's kind of doing an action to make the world fit more her model and they they do another example afterwards but I I'm kind of curious about exactly how this tradeoff is done when would you choose to do one thing over the other when would you choose to do an action versus when would you choose to do a um like an update of your internal model uh especially in a situation where you don't know you know perhaps if you've had this situation happen many times you kind of learned that it's important to know that it's a frog uh so you don't want to look away you want to up your model but what if you're meeting it the first time I was a little bit confused about that great question anyone have a thought on this you're likely doing both in parallel much of the time somebody who's too fixated on one or the other that's tends to be a neurotic problem or Worse what that's greate um but there a few few ways to take this first there's the secret third option which is neither discrepancy doesn't have to be resolved but it is kind of like the equivalent of a a ball on a shelf it's like an unresolved potential energy or like an unburnt candle it's like an unresolved Gibs free energy differential however it isn't the case that a discrepancy is resolved one of these two ways however these are the two ways that there can be the reduction or of discrepancy or or discrepancy continue to increase so then the question is like okay in in what ways uh are or could be discrepancy resolved through inference and through through action um there's defin ways to take this um in the in the particular partition so that's splitting up into these four nodes these two nodes that constitute the the two directions of the blanket and then the internal and the external states that are partitioned off from each other from that blanket those are all happening simultaneously so one answer is they're both separately in play so you could have a situation where where neither uh belief updating nor action updating were happening like a rock you could have a situation where both were concurrently um updating sometimes it can be helpful to think about these kind of multiple time scales of what changing the mind can look like the fastest time scale which is kind of equivalent to the forward pass on a neural network is just the the forward inference that's just all the kind of parameters are staying the same and then new observation comes in basy and update cranks then at a slightly slower time scales um corresponding to attention and learning mechanisms there's kind of like those are corresponding to like weight updating in a neural network and so those those are like slower kinds of changing the mind um changing the world through action also like that's very dependent on what the kind of action is for example here the um eyes were moved so it's a muscular action to bring the observations into alignment with the expectations however there wasn't a change in the world so visual activity is kind of a special kind of action because it doesn't modify the world but it is still a way to reduce discrepancy through action so it's kind of a kind of on the borderline like is that changing the world me your eye is part of the world so that is but it it doesn't modify like any kind of latent State then there are the actions that have like a more you know turning on the air conditioner or the heater that actually modify the world so short answer they're both happening uncoupled and then when you look at the specifics of the generative model then you can you could separately talk about like the pro cities of a model to update its beliefs if it has a fixed belief distribution then it's not going to update its beliefs if it has a very broad belief distribution prior then observations May update its beliefs a lot and then similarly depending on like the efficacy and the agency of the actions that are being discussed then maybe taking actions to modify you know changing the temperature in the room if you have a really fixed physiological prior on body temperature and the air conditioner is right there then that's going to be reflected in the model and you'll see changing the world if there's no air conditioner or the agent doesn't know about it and it's able to like learn different physiological um temperatures or has a very broad temperature range that it could be okay in maybe that would be the behavior that you'd see here's like the temperature simply changing one's mind does not seem very adaptive but acting to lower the body temperature is right thanks a lot and I think I was just interested in about um so that makes sense to us that it uh that we get more out of CH trying to actively change our body temperature than just believing oh um I actually don't have this body temperature anymore but where are we learning that is that through some kind of trial and error that we're learning this is this like an evolutionary prior or where did we get this from yeah it's a it's a good question so um when you construct the model then you get to create whatever the distribution is then ask you well but how how did that prior on body temperature come to be and that the evolutionary account is um you know in the 98° thermal bacterial niche the bacteria that didn't have optimal growth that didn't expect themselves to be at that temperature those have been swept off the table so we end up seeing a phenotypic distribution that's the the fitness distribution of the bacteria in the thermal vent that corresponds like to a thermal gradient um another another kind of consideration is actions are being selected also from a probability distribution like there's no proposal of some auxilary reward distribution we're updating our policy prior habit into the policy posterior we'll look at that like later in the chapter so we're doing inference on action we have inference on perception those the that's active inference um so in terms of the belief Distribution on perceptions that you can think okay like the beliefs about the temperature in the room but also there's the beliefs about turning on the air conditioner so now which which um then the the um sort of liability of both is defined so if the distribution that one of these if the Habit distribution is like um 100% I turn it on 0% heater you know 100% air conditioner 0% heater then that is like it's like terministic action selection versus one that's more spread out across the action possibilities so you can kind of get that interpretability when you see the model and then that's why there's like when you do specify model it's like and this comes up in chapter six you kind of think more structurally about the model and then there tends to be a lot of parameter sweeping and like investigation across different combinations like you could try 10 different um kind of widths of Prior on the perception side from very tight to very loose and then similarly on the action side sweep across 10 by1 and then look at um you know color in each cell by the probability of times that they turned on the air conditioner so it's it's not always just about finding like one kind of specific parameterization especially with these more intuition pump models but often it's more about sweeping and then looking at like oh um you know there's kind of a there's a manifold where on this side this is the case and on that side this is the case thanks a lot yeah thank you um any other question or chat oh also just to sort of yeah yeah go ahead Andrew are you saying this but the space could potentially be not smooth right if there are if there are Jagged jumps in terms of response to different probabilities does does I'm not trying to be too technical but does that make sense it it could be a a jagged space that has like piecewise squares or something yeah it could be a fully discrete space I mean there there could be a continuous space Also with with nonlinearities but also it could just be a fully nonlinear or I mean a fully discrete space like it could be a checker game or a chess game so then that then that um then those sweeps would have sharp boundaries okay look's yeah go ahead oh that just matches that it doesn't have to be nice cuz I was thinking why would you do parameter sweeping instead of some type of optimization like but there may not be a nice you know smooth space to do optimization on so yeah yeah I mean you you've explored that and know a lot of areas sometimes when you fully like the fully synthetic when you fully get to specify it like we're playing Connect Four so it's a discrete space there's a discrete action space there's a discrete game playay space or tic-tac-toe then it's like well behaved and discret but then when you get into like discretized forms of systems that have continuous character then like then there's all these other issues that could come up also just want to make one kind of General comment just chapters one and 10 are are like book ends it's like introduction and kind of conclusion and they're they're similar chapters they contextualize active inference chapter four is where we're going to meet the actual generative model itself the generative model is kind of like the transfer station or the midpoint on the high road and the low road we're in chapter two so we're talking about model architecture so these are things that you can build these are ba go starting from Bas equation and using basian statistics to build up architectures that have a perception inbound sense making element and an act outgoing control theoretic element the high road is where we'll get to the free energy principle it's kind of like imperative for survival and some of those topics but in the low road it's kind of like tinkering and cobbling with different model architectures and motifs I have a question uhhuh so I'm from the transitioning from uh free energy to expected free energy um I mean I I guess it makes a lot of sense to to actually take the expected value of the the free energy but is there like that derivation wasn't really that clear to me it was just like here's the expected free energy do you know have any good resources or where you can actually see that derivation great question well one is in octopus's math work um I I think tomorrow and and continuing it'll be drawn out um a little bit also over it you can and you can follow like click on the little icon for the a here's equation 2.5 so few few things about the equation um we have the screenshot also there's there's the uh plane latac this can help if you are putting it into like a language model and we have some of these experiments with interpreting and asking questions about it not not saying they're all

100% like complete or anything but in the canvas for two point and then also for for many but not all the equations there's the um natural language pronunciation basically which can be helpful and so if it's missing people people can add it so that that can um we we'll have to fix this because it looks like a lot oh I I kind I know what happened here okay there we go so these are the lines all right in 2.5 there's a few resources so um let's just where do how do we get to 2.5 and 2.6 these are some of the key equations okay 2.5 equation 2.5 is variational free energy it these are both vfe and EF are functionals so there are functions that have the arguments that are other functions that the arguments can include other functions so f is the functional Q is the variational belief distribution Q is the distribution that we get to control the form of so it is set by the model to have like an analytically tractable form like a gaussian approximation so even if the world is non you know the world's bodal and then the model saying but but I'm using a gaussian so that gives me all these like nice methods then also all the pathologies like if the world's biotal and you're fitting a gaussian you could you know potentially you fit to only one hump but not the other or you end up blurring over both the middle so it's not not not that if you know if there's a mismatch on form then this will lead you downhill to not necessarily like the best of all possible outcomes but given the selection of the form of Q it will be the best possible outcome it's a functional of the belief distribution and one incoming data point why so variational free energy is kind of like this real time check on like how surprised am I with this data point variational free energy bounds surprise minimizing surprise is equivalent to maximizing model evidence if you were minimally surprised with the new temperature readings coming in that would be equivalent to saying you had like the maximum likelihood model of the temperatures coming in you're making the best possible predictions is equivalent to saying that you were minimally surprised so this is bounding minimal surprise then um it is only dealing with belief in a moment and one data point coming in so that's why it's this kind of like real time sense making indicator 2.6 is when we start looking into talking about observations that haven't happened yet at the future and action planning G expected free energy it's taking in pi which is this policy vector it sums to one it's a probability distribution over actions the policy prior is the Habit distribution G is going to sharpen that policy prior into a policy posterior which then you could sample from like according to the probabilities or you could just select the likeliest action okay how do we get between the different lines again in this canvas here for 2.5 and 2.6 there's some useful resources um um here's equation 2.5 walk through um Ali ramju Yak smal Jonathan shock have all done like have have contributed variously to these sections so that's one good PDF um here I think this was yakups the interesting thing for 2.5 and and this is like again why there's so many dots to connect with the math is 2.5 actually um it's not that one you go to two and two you go to three maybe there's a maybe there's a way to do that as well but here's the derivation going from one to two and then one to three so these may be some useful resources also here let's see if this resource is still there Professor Jonathan shock has written up a sort of math overview August 12 I don't know if that renders to the current date just or if it was like actually edited at this current date that also has a lot of information so that's something that that takes a lot of uh coats of paint I think but it it it really is there and grounded and if we can pull out and help highlight like all those threads and show okay here's where this is grounded in the definition of Shame and entropy or definition of surprise and likelihood that'll be cool one other thing I'll say is um variational free energy is kind of more of a classic quantity it's the quantity that's used in variational inference um like variational physics variational Auto encoders and so on whereas expected free energy is just one choice of a way to model futures for example there's also this is EF expected free energy also there's free energy of the expected future so there's several other future oriented approximators that have been proposed but variational free energy is kind of the classic Adam yeah like something that I'm wondering about and I haven't really quite like understood yet um is that so from like my my own personal past and like where I'm coming from up until active inference I had been reading quite a bit about like maximum entropy inference in like John Hart from Berkeley and Edward Thomas James and uh even like pier llas and um trying to think so anyways definitely like ET James he was really like a proponent of Max ant inference and um prior to active inference I had I had been thinking about like the stability of syst systems in the maximum entropy State and the way that if if the energy of the of the free energy equation is is constant then by the increase of entropy it automatically minimizes the free the free energy right so just that the Min the men free energy is the equivalent or not necessarily but can be equivalent to the maximum entropy and so I don't know if anyone else has been thinking about this or maybe if if you like also I don't know any thoughts on that yeah that's a great question it it H has not been um fully fully fleshed out but here's one kind of recent discussion where it came up and where they explicit ly highlighted this okay so here was it it um okay this is the octave data sampling paper so observations um it's kind of this is kind of like like building on that discussion of the equation 2.5 VF so that's where the variational free energy for data point and a prior are going to be described you could also look at a data point and and your prior and disc like what was the epistemic value of that data point coming in how much did we learn from it um here they're in a clinical study setting so like the real Target uncertainty to resolve is about the efficacy of these different treatments um they use max ENT sampling and then they also compare that with the expected Information Gain driven sampling so there's a very um okay here we go several examples would yield identical results had we adopted data sampling based upon Max entropy sampling the key differences emerge when variance around predicted outcomes is in homogeneous so when there's equal variability profiles like let's just say that we're we're interested in the temperature distribution in this room and all of our sensors have equal VAR sensor variability then we can sample from the most the sections that we're least certain about the temperature and that's like a Max ENT strategy however let's just say that there was regions in the room that had different variability and there were sensors with different variability so just to kind of simplify let's just say we had high and low variability sensors and there's high and low variability areas in the room so if we were just going to sample based upon variability period measurement variability like total variability then maybe this we'd be ending up sampling from sensors that were highly variable but regions where we actually were not not that uncertain about the temperature at that location so in the case where the sensors where the sensor variability we don't have to worry about then these approaches are equivalence but there's actually an additional level of nuance possible so it's it's closely aligned with Max and I think again it would be it would be great to to draw this out better but um it's sampling from a total epistemic value which is able to kind of like accommodate that case where there's variability sensors and the variability of the temperature parts of the rim yeah I see that I'll have to look into that that sounds interesting cool yeah the the slides are there the and the paper um and and we we that's an interesting thing to keep on exploring what exactly what exactly um this is just on the website I can get to this thep yeah um just here's the just like 57.0 that was the the background and then 57.1 and point2 um Carl friston and Thomas Parr joined I forget which one of the weather was d one or. two but in one of them Carl gave a a very vifer defense of epistemic value because often times people say Okay epistemic value learning in service of pragmatic value right like learning is good for something when it's pragmatic and he gives a extended thought on like how they're kind of separate but they're unified under the expected free energy but um like epistemic value is its own distinct kind of value from pragmatic value whether or not those learnings contribute to epistemic or I mean contribute to pragmatic outcomes I don't no sorry all over my hand okay yeah th this the epistemic um like active data sampling like on the biological side it's really interesting because it describes for example is secades epistemic driven action selection but then also they explore having a cost function so that could be like how much how much cents does it cost to make an API call and then that is the parameterization question is like how much epistemic value how many bits of information are worth one cent of an API call um or in the case of the clinical trial there's like you could have a cost associated with the number of times that sampling occurs or um cost reflected by a preference for survival but if the preference is purely for survival and it was like some kind of extreme utility driven clinical trial then a sampling algorithm would choose to sample nothing so it' be uninformative with zero observations however it would have achieved maximum preference satisfaction because it wouldn't have observed any sickness or death so they talk about a few of these kind of like interesting cases and they they build it up very clearly like how do you add in the cost and all these different things and this is like a different graphical model than comes up in the textbook again it's the clinical trials setting but but to connect it with the textbook these are basing graphical models and so a lot of the the kind of structural analysis of a situation comes down to what are the variables what are like the observable variables what are the environmental variables what are the and there's there's not really like an agent like kind of sense an action agent unless you chose to see like the clinical design team's observations and decisions in in that light um what are the observations what are the environmental variables what are their influences upon each other that lays out like the skeleton and then there's the question of either generating synthetic data from that skeleton in that in the forward direction or explore in chapter n of the textbook is like how do you go from observations to inference on those latent factors like um what is the efficacy of the treatment anyone else give like a thought on chapter 2 or question or a page or we can look at some of the questions come on go for it yeah thanks um yeah I've been thinking a bit about like the distinction between pre energy and expected pre energy and sort of thinking about like so in like in a way aren't all actions requires planning or like is there actually a clean distinction in practice when we when we minimize F and when we minimize G or is it like a bit of a fussy line in between and in practice we're only doing the one or if you know what I'm getting at yeah great great question so let let's go to pmdp documentation that this can this can um clarify a few of those things one uh heris stic is if you're considering explicit planning so more than one time Step Ahead like policy sequences that have a Time Horizon that's not one if you want to do explicit planning Tre search policy roll out um deep active inference then you do need to use expected free energy so if explicit planning is going to come into play then you go with expected free energy if it's just single time step then it can you you can just sort of have that reactive updating um yeah hon let me um that's in in the uh equations equation 2.5 put in the chat yeah so if there's explicit planning yes EF has to come into play if you have an agent that is not doing explicit planning it could do action selection using vfe so what that would look like and and is like you have um a belief Distribution on the temperature in the room and a belief Distribution on your habits related to the air condition conditioner like I'm 50% likely to turn on the air conditioner or not new observation comes in it instantaneously updates the belief distribution that's the sensemaking element and it instantaneously updates your um your policy distribution which is then sampled from so there's not roll out or anticipation explicitly of future unobserved outcomes which is what you would get with ef but you only get one time step there um here in in pmdp it's um they really especially in the um documentation make it pretty clear and declarative like there's only two actions here but here even for a single time step moving left and moving right they're they are considered with expected free energies because there's actually like a question of what what would I expect to see if I went left and what would I expect to see if I went right so that's kind of like you know there's a lot of way ways to shade that but that's kind of like the theological aspect of action is when expected free energy comes into play which is like what what would I see if I went left versus what would I see if I went right right versus the sort of past pushing forward without an explicit teeology would be like I have a 50% chance of left and right prior on policy I just saw um a light or a glucose gradient from the left that just bumped me to 6040 I sample from the 6040 and I go left but it wasn't based around what would I see so it wasn't like based around that what what is the epistemic value and the pragmatic value of these two Alternatives that's the real policy like planning as inference versus more of like a reactive policy just habit unupdated or updated habit distribution and then just drawing from that without explicit consideration of the consequences and maybe just a short followup question um so yeah I guess you can make it like really clear when when writing a a program about it but in the context of of living organisms when you yeah so like what is what is one time step then um could maybe like develop that discussion a bit on like specific for living organisms [Music] um yeah well math territory H how do we develop continuous Andor discrete time models maps of complex multiscale systems like there's not really a natural time click for a body per se it's not simply like the heartbeat it's not the seconds it's not the breath so then there's a lot of Moder degrees of freedom with how they model time um and then to what extent the modeler chooses to test here's some models I made where there's planning happening and here's some models where planning isn't happening that that that's one thought but but yeah like does anything plan or when does planning happen or like one old active joke was like active inference only works for one time step but then you can compress and make that one time step very complex so that you like it could be one mult like one nested time step that that would subsume what looks planning like if if um let's just say that we have we were um talking about like financial decision making or something like that and we had a a a policy decision that was being made at the yearly time scale if someone only knew about day-to-day decision making that one time scale of the yearly model might look like it had been planned over many days or there could be a map that just said we have a single policy we make one decision per year but then that model wouldn't be able to make decisions day by day but it might look you know it this just gets into like how I mean how and we can make many models from the biological phenomena and then that that's kind of the interesting question like is there evidence for planning in is secades or is it simply marcovia and we can totally account for is secades based upon location of the Gaye at a moment the incoming observation and the belief distribution about what would be epistemically valuable or like is there a residual that that model can is not accounting for that is suggestive of like twostep Cade Planet I don't know but those are like the kinds of of research Pathways where where if you are looking for where is their planning happening thank you um actually sure so uh I'm pretty new to active inference and I'm kind of trying to right now get a intuitive feeling for some of the um equations especially the one we were looking at before I'm wondering if we could just talk a little bit about how the terms kind of relate to each other and how actually I might be able to send you uh I just kind of cooked up something quickly before here in the chat I kind of tried to create a visualization of the um I don't know if you have access to it uh I sent the link Yeah so basically let's see here yeah so the idea is that you have like the Q of X and the P of X and Y and as far as I understand I might have made some mistakes but as far as I understand you minimize the free energy we're looking at like the top uh version of the equation right the first line um so we have the Q ofx which is our belief distribution is that correct like our belief of the Hidden States yep then we have the P of Y and X which is just the probability of Y and X happening as far as I understand when P of y and x and q of X are like exactly the same that minimizes fre eny is that correct yeah well great work with this this is this is super cool and it'd be awesome to to continue to develop these like interactive slider based things which which many people have kind of like brought up but um yeah like what variational optimization does is the modeler has chosen their family of variational distributions that they're going to be sliding the well- behaved knobs on so like we're going to be using a gaussian again that's not saying the world is gaussian it's saying the the modeler is choosing to to do optimization you know descent with this G um and then the ball rolling downhill on that variational free energy landscape until it rests at the bottom what is the bottom of that energy well it's where that observation coming in given the belief distribution is is the least surprising possible or it it bounds surprise now that bound if the generative process were gaussian and the generative models gaussian then you'll get like then it to it bound surprise and and kisses it if the generative process were something different and you were doing a gine approximation you still would get the best possible gussian approximator but then there would be still like a constant that that was like surprised that couldn't be reduced because you were fitting the best possible Galaxy but it still was being um because it was like not the same family as a generative process couldn't reduce all the uncertainty but yeah this is really cool right like this is just a very quick um but but I do think it helps to kind of play around with it and get a feeling for how things change because one thing I read in the book was that if you set the mean of both to zero um you can see that uh have to add some kind of quick reset button as well but you can see that uh as the they say that if you have little evidence which would mean that your P of X and Y would be really really um kind of broad right so if you turn the the bottom uh all the way up like the yeah you you turn it up up up up up then you start getting a free energy that's higher but then actually as you they say that you should as far as I understand it that when you don't have a lot of evidence you should also be very uncertain certain so if you I'm seeing the P of uh X and Y distribution as being broad as not having a lot of evidence and then if you take the Q of X and take the standard deviation up on that while the uh or sorry the standard deviation you'll see that if the standard deviation of the PX and Y is also very very broad it does minimize uh variational free energy as well does that uh make sense with what what the truth is or yeah like I'm I'm really interested in this um change from a uh single well to potentially I mean potentially some plateau and then here potentially like a symmetry break where if the first observation coming in were here maybe it breaks to the right and then continues to follow the gradient and down to here and then a future observation couldn't even if it came from here it'd be like okay now back to here and back so then it's like trapped in an energy well so it's like what are these situations where even when we have a gaan prior and a gaussian generative process like what is happening where we're we're getting a more like a um quartic like kind of two energy well free energy distribution so I'm not sure if I'm illustrating it in the wrong way and maybe uh like the blue graph is basically Q ofx uh times the natural logarithm of P of Y and X before it gets kind of summed up into the so that's kind of what that's showing I don't know if that was the same thing you were referring to but just so you know yeah cool maybe octopus and others can can play with it and keep keep on developing it or um or post the the repo that it comes from but this looks cool um so just one quick followup so earlier in the beginning you were talking about I mean can I understand it in that way that when you're very certain because they say in the book that if you are if you have little evidence then you should also not be so certain in your model um and I'm wondering whether that uncertainty is characterized by having a very broad Goa for example and whether the lack of evidence is also uh kind of how do you say represented by having a very broad gion if you're using a go yes this kind of relates like you have to the standard deviation um what comes up later is this like learning by counting so let's just say we've observed one heads and one Tails we might have um a distribution belief what is the probability of this coin coming up heads it's centered at 0.5 but it might be very broad whereas if we had one million heads and one million Tails observed then we would have very sharp Gan around 0.5 so there's different ways to kind of implement that but but broadly when you're less confident when there's more epistemic value on the table then belief distributions come to look like uniformity and then as you gain Precision it gets more of a central tendency and then the extreme case is basically a dir Delta function so it's just a spike at one location that might not even be learnable and then it's just like it's here and even if you got even if you have a spike at 0. five and you it just then it doesn't matter what you observe because there's no probability Mass on outside of that one Spike so it's like those observations are are like discarded whereas the uniform prior um is which is kind of like the frequentist case interestingly is like the the I mean it's the least bias in terms of its sensitivity but then then you realize oh but then the but the updating is the bias that is statistical bias right cool thanks a lot and if anyone else is interested in like building these kind of tools or something please reach out to me because I think it' be pretty nice to have some more visualizations of how these things play together so but thanks a lot for the explanation cool yeah it'd be so fun like for each equation just like even to like we you could have like natural language and translations pseudo code some different languages some interactables like there's what are the kind of like different forms that that we could interact with and then also like where and when and how like in theory and in the specific implementations you know the dozens of implementations like where did the modelers choose to use the energy minus entropy kind of statistical mechanics decomposition of f where did they use a complexity minus accuracy kind of Statistics basing information Criterion type and then where is there like a Divergence minus evidence like if they're equivalent then does it matter or like are one of these more computationally intensive to compute or is there like is there a reason or a setting why one of these decompositions on F is better and then that really gets opened up in the policy selection for expected free energy like we talk the most about epistemic plus pragmatic value that's a very interesting decomposition for for policy selection but it's equivalent to these two ambiguity plus risk kind of classic control theory esque um investing type decompositions and then both of these are equivalent to the again energy minus entropy but then it's like but then are we even talking about planning if we're talking about energy minus entropy or how how and then like what if we had like a vent diagram it's like here's the total amounts and here's the the variance that's getting partitioned into epistemic here's the variance that's getting partitioned into pragmatic now I don't think the same this I don't think the same variance is going to ambiguity and risk nor to energy entropy because the terms are not equivalent to epistemic plus pragmatic value so it's like so what loads on pragmatic value then does that get does that get you know loaded onto risk or onto ambiguity does pragmatic value get loaded onto entropy or onto energy again is there a reason or how did different modelers do it the top line for G is is the most common and the top line for for f is the most common I don't know though but those are like they're they're given as a first line because they're kind of like the classic form but it's it's an interesting topic the rule of four or more octopus yeah what is the rule the rule of four is an idea from teaching math that developed in the 90s that was like every concept you look at you try to have at least four different representations usually they're verbal numeric symbolic and um verbal numeric symbolic and visual and uh what what you were saying about like we have all these different representations and we could add more was just echoed that for me cool yeah and here we have we have there's we have different words in Lang different natural languages different graphical um symb it could include executable code and then there could be some some wild cards games and stuff like that but yeah it always pops up it seems it be I mean for each of the equations we could have a column for these and then that creates like you know if there's 50 equations then it's just like we can just say okay here's the empty cells let's try to get something into each of these cells CS yeah that's I'll just delete this because SE um yeah I mean this is cool and these are great questions that's what chapter 2 again we haven't even gotten to active inference this is saying okay op um Bas optimality probabilistic probab probably approximate basee like pack um uh probably approximately correct here's the par partition splitting systems into these four nodes minimum of four that's another rule of four or more kind of splitting into four two blanket States internal and external getting into what's basically a variational auto encoder here it is is framed in the Sens making setting and then extending that variational autoencoder into the decision- making setting showing that that expected free energy unified objective or or imperative or kind of like loss function a lot in common with with loss functions that the special cases of expected free energy fall out as other well-known pis stics like if you only consider pragmatic value you get expected utility Theory no epistemic value nothing to learn special case of G expected utility in contrast if all outcomes are equally preferable so there's no pragmatic value on the table then epistemic value yields Infomax so it's kind of like saying if you're going to compose with these basy in Legos either in the the real time inference and then also extending out into the the planning and the action selection like anything any any any brick structures you build on the low road are going to you're going to be able to approximate them with variational basian inference chapter 3 is going to come in with with more of this phys flow based model and and say and um for the organism like observing itself or for some for you observing something else something will act as if it is bounding its surprise about something else so low roads like saying anything you build you're going to be able to approximate it do variational inference with that surprise bounding and then here is more of the the physics saying and you can model things such that it that it looks like that's what they're doing and then in four that's where the generative models actually get shown and it's like this is this was built up on the low road and it's kind of compatible from the high road and that's why it's placed at the middle it's just that also four has like again some more mathematical background like Jensen's inequality and properties logarithms um variables and their dependencies in graphical models but then figure 4.3 does bring us to the disree and the continuous time generative models so that's why it's always just kind of like like many codes of paint like just kind of reading through because there's not really just one order especially as everyone's going to have different like questions about different parts um so that's kind of fun like what are the more the more pure background materials like putting this as figure 4.1 might make it seem like this is like finally we got to act in it's figure 4.1 it's like well no maybe that that's just a generic property of logarithms cool um well so uh next week for S we uh it'll be later or join at this time next week and uh either Andrew or I will revisit applying active again there's no there's no specific order so that that can be helpful like chapter 7 goes into essentially the top half of figure 4.3 so may be useful to to see it and um we can go into pmdp python based very imperative pseudo code like form uh format for the programming like get the observation update this do that calculate this do that um or we could explore a little bit towards RX and fur which has this declarative style where it's more like you declare the base graph full stop then you declare a kind of like logistical scheduling of how how do you update different parts of it so that has there's a few more steps that are possibly initially unintuitive but then in operation it gives the RX infer style of declarative programming a lot more performance and flexibility okay cool any I mean any with anything else in the times between just make make the notes on the chapter Pages or just prepare them however oh I'll copy and Dave's on note ruer four in toolic linguistics they don't write abstracts like they used to okay thank you see you all later bye